

Boarding Sensation Presentation of the Biped Walking Robot with a Low-cost Two-axis Motion Platform

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ABSTRACT

We render a boarding sensation of a biped robot at low cost and high immersive by approximate the 6-DOF motion such as the impact, vibration, and steep slope experienced on boarding a biped robot to a 2-DOF rolling motion at max ± 25 degrees in both at pitch and roll directions with our low-cost two-axis motion platform.

Keywords: VR application, motion platform, boarding sensation, biped walking robot.

Index Terms: Human-centered computing—Human computer interaction (HCI) —Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI) —Interaction devices—Haptic devices

1 INTRODUCTION

In the field of virtual reality (VR), motion platforms with swingable seats are used to express the feeling of boarding a car, airplane, robot, etc. As a previous works on boarding sensation presentation of the robot with a motion platform, Zordan et al. expressed the vibration of the robot walking by swinging about ± 7 degrees in the pitch direction and reported that resulted in the feeling of walking by demonstration at the SIGGRAPH 2016 VR Village [1]. However, they have not yet led to achieve representation high-frequency vibration caused by the impact force of the foot and the ground while the robot is walking and make the robot walks on fields with large inclines because it is structurally difficult to achieve a large swing on previous motion platform with a low degree of freedom.

Therefore, in this work, firstly we model the vibration caused by the impact force of the foot and the ground while the robot is walking. Then, we approximate the 6-DOF motion experienced on boarding a biped robot to a 2-DOF rolling motion at max ± 25 degrees in both at pitch and roll directions with our low-cost two-axis motion platform [2]. Finally, we describe implemented result and following capability result, we consider whether the vibration caused by the impulse force of the foot and the ground while the robot is walking can be expressed on our motion platform.

2 SYSTEM OVERVIEW

As shown in Figure 1, we use two joysticks for both hands to operate a virtual robot in the Unreal Engine 4 (UE4). By inputting two joysticks in the forward (Y-axis) direction at the same time, the robot starts walking forward, and if one of the joysticks is tilted forward and the other is tilted backward, the robot swivel. At this

time, the cockpit moves up, down, left, and right based on the preset keyframe animation. According to the movement of this cockpit, the pitch and roll angle of our motion platform, the expansion / contraction speed / acceleration of the actuator are calculated (about 120Hz). These parameters are used to control the amount of expansion and contraction of the left and right actuators of our motion platform (about 30Hz) [2].

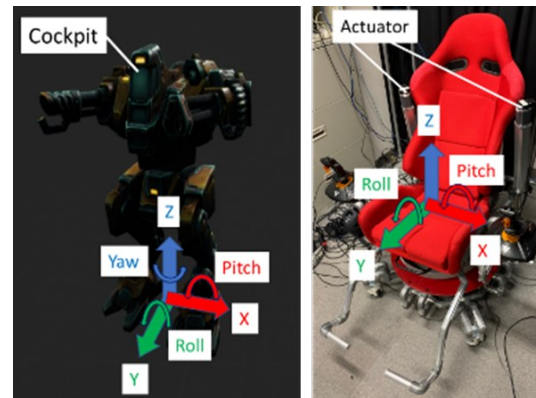


Figure 1: Overview of a virtual robot and our motion platform.

3 BOARDING SENSATION PRESENTATION WITH A LOW-COST TWO-AXIS MOTION PLATFORM

Since our motion platform rolls in the roll and pitch directions, it does not actually move up and down. Therefore, in this work, the vertical movement of the Z coordinate value of the cockpit is replace with rolling in the pitch direction. The vertical movement Z [cm] of the cockpit is expressed by the following formula.

$$Z = 1.76 \sin\left(\frac{2\pi}{0.5335}t - 0.029\right) \quad (1)$$

Then, the impact force when the robot's foot lands on the ground is approximated by the spring damper. The equation of motion is expressed by the following equation.

$$m \frac{d^2 Z'}{dt^2} + c \frac{dZ'}{dt} + kZ' = F \quad (2)$$

Where, Z' [cm] is the displacement, m [kg] is the mass, c is the damping coefficient of the damper, and k is the spring constant. F is impulse force.

Since the impulse force works for a very short time, the impulse force is zero immediately and the displacement remains almost unchanged. Therefore, after the impulse force become zero, the equation of motion is as below.

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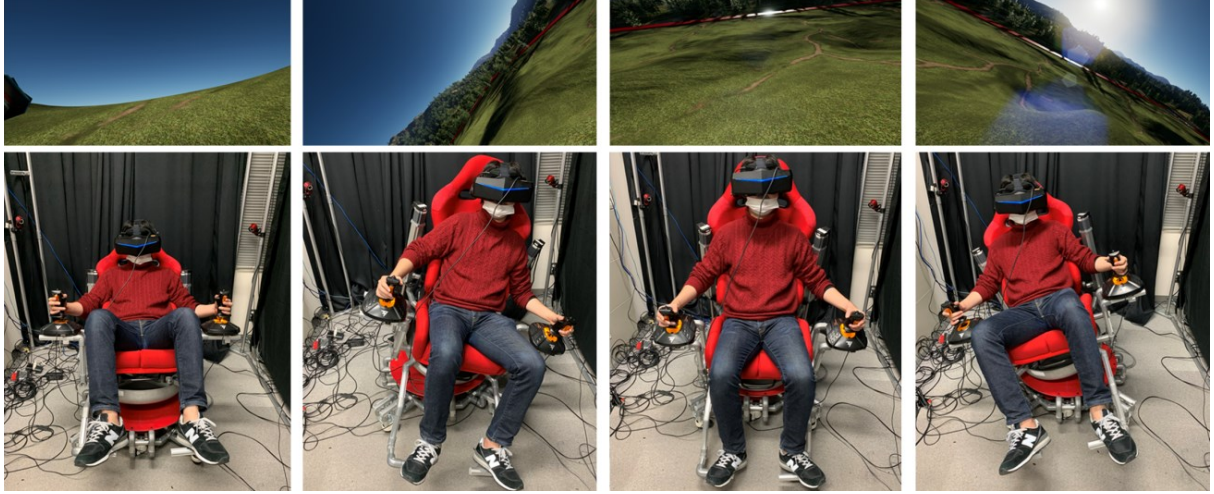


Figure 2: Rolling states of our motion platform while operating the virtual robot.

$$\frac{d^2 Z'}{dt^2} + 2C' \frac{dZ'}{dt} + (K')^2 Z' = 0 \quad (3)$$

Where, $C' = \frac{c}{2m}$, $K' = \sqrt{\frac{k}{m}}$.

Here, solution of eq. 3 is as bellow.

$$Z'(t) = z_0 e^{-(K')t} \left(\cos \omega t + \frac{k}{\omega} \sin \omega t \right) \quad (4)$$

Where, $Z'(0) = z_0$, $\omega = \sqrt{(K')^2 - (C')^2}$.

Finally, the amount of expansion and contraction of the actuator is controlled at an appropriate timing based on the calculated value $Z'(t)$, and is replaced by rolling in the pitch direction. In this work, z_0 , K' , C' are set to 1, 70, 10 for now, and these parameters adjustment is future work. In addition, tip the robot so that it is perpendicular to the ground (see Figure 1).

4 IMPLEMENTED RESULTS

Figure 2, 3 shows implemented results. As shown in Figure 2, 3, it can be seen that the seat swings greatly according to the inclination of the ground. However, as shown in Figure 3, it can be seen that the followability of high-frequency vibration is weak. Therefore, as future work, we will adjust the parameters, z_0 , K' , C' , and speed and acceleration of the actuators.

5 CONCLUSION AND FUTURE WORK

In this work, firstly we modeled the vibration caused by the impact force of the foot and the ground while the robot is walking. Then, we approximated the 6-DOF motion experienced on boarding a biped robot to a 2-DOF rolling motion with our motion platform. Finally, we described implemented result and following capability result, we consider whether the vibration caused by the impulse force of the foot and the ground while the robot is walking can be expressed on our motion platform. Currently, the seat of the robot swings greatly according to the inclination of the ground, but followability of high-frequency vibration is weak. Therefore, as future work, we will adjust the parameters, z_0 , K' , C' , and speed and acceleration of the actuators.

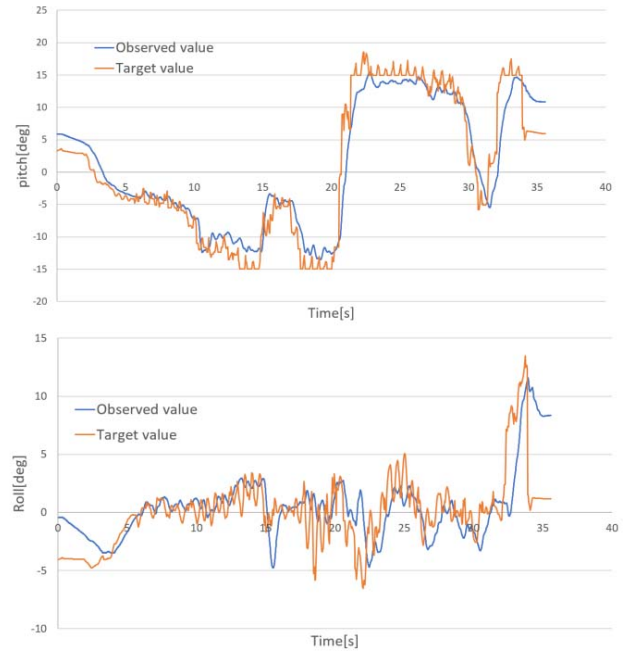


Figure 3: Target and measured values in pitch and roll angle.

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